

BONSAI: Adaptive Task-Graph Functional Replay for Low-Forgetting Continual Learning

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Abstract—Continual learners must acquire new concepts without allowing later updates to overwrite previously useful input-to-output functions. We present a new BONSAI variant, Adaptive Task-Graph Functional Replay (ATGFR), that combines a variational information bottleneck, cached task-graph relations, hierarchical retrieval, a bounded local positive-definite metric, shared low-rank task adapters, and function-level replay. ATGFR keeps a class-balanced coreset from training inputs only, stores the completed task’s logits and latent features, weights replay by task-graph similarity, and increases replay pressure only when measured old-feature drift exceeds a budget. This design directly targets the failure mode of parameter-only consolidation: stable coordinates do not guarantee a stable function. On a four-task, twelve-class structured-image stream evaluated over five fixed seeds, the matched diagonal control obtains 0.1000 task-aware accuracy and 0.3778 peak-before-final forgetting, whereas ATGFR obtains 0.6333 accuracy and -0.2000 forgetting after mass-preserving relation normalization. Task-free accuracy is 0.6333, with 0.5667 mean task-route accuracy and 0.498 times model-parameter memory. A predeclared robustness grid spanning two and eight tasks and eight and 128 input features improves in all four cells. We additionally evaluate ER, DER++, ER-ACE, and ATGFR on Split-CIFAR-100 using one identical compact convolutional backbone, exemplar count, optimizer, and three predeclared task orders (two seeds per order). ATGFR reaches 0.0358 accuracy versus 0.0321 for ER, with 0.0037 forgetting versus 0.0038 for ER, but remains far from a strong CIFAR-100 classifier under the deliberately small training budget. A factorial replay ablation, wall-clock accounting, and task-order sensitivity analysis expose which components are and are not supported by the data.

Index Terms—continual learning, catastrophic forgetting, functional replay, task graphs, knowledge distillation, parameter-efficient adaptation, information bottleneck

I. INTRODUCTION

Standard neural-network training assumes that data can be shuffled, revisited, and jointly optimized. Continual learning instead presents a sequence of tasks, domains, or classes and asks one learner to remain plastic while preserving prior competence. Sequential optimization can overwrite useful representations and decision boundaries, producing catastrophic interference [1]. This stability–plasticity problem is particularly important in deployed systems, where replay may be restricted by privacy, memory, bandwidth, or data-retention policy.

Existing families attack different parts of the problem. Parameter regularizers such as Elastic Weight Consolidation (EWC) and Synaptic Intelligence (SI) protect important coordinates [2], [3]. Gradient constraints and subspace memories

retain directions that should not be changed [4], [5]. Parameter isolation methods such as PACKNET and Progressive Neural Networks (PNN) freeze capacity or grow new columns [6], [7]. Replay methods preserve examples, logits, or generated samples [8], [9], [10], [11]. These methods expose a practical gap: parameter stability, gradient stability, and function stability are related but not equivalent.

This paper introduces ATGFR, a system-level method for the next modular BONSAI architecture. Its central design choice is to preserve a bounded sampled approximation of the old function, rather than relying only on a parameter point or a gradient subspace. The replay signal is not uniform: the cached task graph supplies a relation weight, while a feature-drift thermostat increases pressure only when the current representation moves beyond a budget. The method therefore spends replay capacity where it is most needed.

The main contributions are:

- 1) A mathematically specified ATGFR mechanism that combines training-only class-balanced coreset replay, logit distillation, latent-feature anchoring, task-graph weighting, and drift-triggered control.
- 2) An end-to-end modular BONSAI implementation with a variational information bottleneck (VIB), cached sliced-Wasserstein and zero-dimensional persistence descriptors, hierarchical retrieval, local SPD geometry, and shared low-rank task adapters.
- 3) A non-cherry-picked evaluation: five seeds for the primary image stream, all four cells of a task/feature robustness grid, repository scaling from 4 to 50 tasks, and a corrected peak-before-final forgetting metric.
- 4) A matched real-image mechanism study on Split-CIFAR-100 against ER, DER++, and ER-ACE, with the same backbone, exemplar count, task orders, optimizer budget, wall-clock accounting, and scalar-memory accounting.
- 5) A factorial ablation that removes labels, logits, features, OT, H0 persistence, Euclidean replacement, and the drift thermostat, with all cells retained even when they weaken the result.
- 6) A quantitative comparison on the current VIB/MLP prediction core against EWC, SI, PACKNET, and PNN, with the same task stream, optimizer budget, seeds, and explicit task-free scope.

The paper’s claim is conditional. ATGFR is better than

the matched new modular controls on the measured stability/accuracy trade-off, and its training-only memory is materially smaller than the diagonal and TGRSC states in the primary stream. The evidence does not establish universal superiority over all continual-learning architectures, nor does it replace a capacity-matched real-image study. The new matched CIFAR-100 mechanism study is intentionally a low-data pilot: it establishes real-image and task-order evidence, but it is not a state-of-the-art accuracy claim.

II. RELATED WORK

A. Continual-learning scenarios and evaluation

Task-incremental, domain-incremental, and class-incremental learning have different information available at training and inference [12]. A task-aware evaluation supplies the correct task path; a task-free evaluation additionally tests task identification. We report both because a model can retain old classifiers yet fail when a deployment router selects the wrong path. Surveys also emphasize that task order, capacity, memory, and tuning protocol strongly affect conclusions [13]. Accordingly, every primary aggregate in this paper uses fixed seeds and the robustness experiment includes every predeclared task/feature cell.

B. Regularization, gradient constraints, and isolation

EWC uses a diagonal Fisher approximation to penalize movement from an old parameter point [2]; SI accumulates parameter importance from loss reduction [3]. GEM constrains new gradients using episodic memories [4], while GPM stores representation-induced gradient subspaces [5]. PACKNET iteratively prunes and freezes task-specific weights [6]; PNN allocates a new column per task [7]. These methods motivate the BONSAI controls TGRSC and ATGTR, but ATGFR addresses a different object: the old function evaluated on representative inputs.

C. Replay, distillation, and parameter-efficient transfer

Experience replay is a direct baseline for reducing forgetting [8]. iCaRL combines exemplars and distillation [10]; DER stores historical logits and provides a strong functional baseline [9]; generative replay trades raw storage for generator error [11]. ATGFR uses the same broad functional-preservation lineage, but contributes adaptive task-graph weighting and a feature-drift thermostat within BONSAI.

The matched real-image study therefore includes ER, DER++, and ER-ACE. ER-ACE uses an asymmetric current-task loss to reduce the pressure of new classes on old representations [14]. FOSTER is a stronger dynamic class-incremental architecture that expands and then compresses features [15]; it is discussed as an architectural reference, while the present controlled study isolates replay mechanisms under equal capacity rather than claiming to reproduce every published leaderboard.

The shared adapter is related to parameter-efficient transfer modules [16]. BONSAI shares its adapter bases and adds only rank-sized coefficients for each task. The VIB component

follows the variational information bottleneck formulation [17]. The task repository uses fixed-projection transport descriptors motivated by Wasserstein computation [18] and an H0 persistence summary derived from a minimum spanning tree, a conservative use of persistent homology [19].

III. BONSAI ARCHITECTURE AND ATGFR

A. Problem definition and metrics

Let the stream contain tasks $\mathcal{D}_1, \dots, \mathcal{D}_T$, with $\mathcal{D}_t = \{(x_{t,n}, y_{t,n})\}_{n=1}^{N_t}$. During task t , the learner updates θ using current data and bounded state retained from $\mathcal{D}_{1:t-1}$. If $a_{k,i}$ is the accuracy on task i after task k , final task-aware accuracy is

$$A_T = \frac{1}{T} \sum_{i=1}^T a_{T,i}. \quad (1)$$

We use peak-before-final forgetting:

$$F_T = \frac{1}{T-1} \sum_{i=1}^{T-1} \left(\max_{k \in \{i, \dots, T-1\}} a_{k,i} - a_{T,i} \right). \quad (2)$$

Negative values are possible when later learning improves an old task beyond all earlier checkpoints. This definition is implemented in `src/utis/metrics.py` and is covered by a regression test.

B. Modular representation and routing

The encoder produces a stochastic bottleneck:

$$\begin{aligned} h &= \text{GELU}(\text{LayerNorm}(W_x x + b_x)), \\ \mu &= W_\mu h + b_\mu, \quad \log \sigma^2 = \text{clip}(W_v h + b_v, [\ell_{\min}, \ell_{\max}]), \\ z &= \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \end{aligned} \quad (3)$$

The KL divergence to the standard normal is added only during training; the deterministic mean is used for inference and repository insertion [17].

For each task, the repository caches a latent prototype, fixed-projection sliced-Wasserstein descriptors, and a zero-dimensional persistence summary. The graph relation between tasks t and u is

$$s(t, u) = \exp \left(-\frac{d_{\text{OT}}(t, u)}{\tau_{\text{OT}}} - \frac{d_{\text{TDA}}(t, u)}{\tau_{\text{TDA}}} \right), \quad (6)$$

and the consolidation weight is

$$w(t, u) = w_{\min} + (1 - w_{\min})s(t, u). \quad (7)$$

Hierarchical coarse retrieval reduces candidate work, after which a local SPD metric and sparse compatibility score rank candidates. The metric has the form

$$G_t = \text{diag}(g_0) + U_g \text{diag}(\delta_t) U_g^\top, \quad g_0 > 0, \quad \delta_t \geq 0, \quad (8)$$

so positive definiteness is guaranteed by construction. The local chart is $\text{Log}_{p_t}(z) = z - p_t$; no global geodesic claim is required.

C. Shared low-rank task adaptation

The adapted latent representation is

$$\tilde{z}_t = z + sU \text{diag}(d_t)V^\top z, \quad (9)$$

where U, V are shared and $d_t \in \mathbb{R}^r$ is task-specific. Once task t is consolidated, d_t is frozen. The adapter itself adds only r scalars per task; repository and consolidation state are reported separately.

D. Consolidation controls

TGRSC stores a rank-limited basis Q_u , normalized spectrum s_u , and mean gradient $\bar{g}_u$ for shared encoder parameters. Its soft retention term is

$$\mathcal{L}_{\text{TGRSC}} = \sum_{u < t} w(t, u) \left\| \text{diag}(s_u)^{1/2} Q_u^\top (\theta - \theta_u) \right\|_2^2. \quad (10)$$

Its signed-conflict projection retains compatible directions.

ATGTR instead projects a proposed gradient g when a first-order old-task constraint is violated. With A_t containing relation-weighted mean and residual directions, it uses the damped active approximation

$$g' = g - A_t^\top \lambda, \quad (A_t A_t^\top + \mu I) \lambda = [A_t g - \epsilon_t]_+. \quad (11)$$

The compact mode limits the retained anchor reservoir.

E. Adaptive Task-Graph Functional Replay

ATGFR keeps up to $m = 8$ class-balanced training examples after each task. For each stored example it retains the label y_u , old logits o_u , and old deterministic feature z_u . During a later task, it minimizes

$$\begin{aligned} \mathcal{L}_{\text{ATGFR}} = \frac{1}{|\mathcal{M}|} \sum_{u < t} w(t, u) h_u & [\mathcal{L}_{\text{CE}}(y_u, f_\theta(x_u)) \\ & + T^2 D_{\text{KL}}(p_u^T \| p_\theta^T) + \gamma \| z_u - z_\theta \|^2] \end{aligned} \quad (12)$$

The temperature is $T = 2$, feature coefficient $\gamma = 0.25$, and relation floor $w_{\min} = 0.35$. The drift thermostat is

$$\delta_u = \frac{\|z_\theta - z_u\|_2^2}{\|z_u\|_2^2 + 10^{-6}}, \quad h_u = \text{clip} \left(1 + k \left[\frac{\delta_u}{b} - 1 \right]_+, 1, h_{\max} \right) \quad (13)$$

with budget $b = 0.05$, gain $k = 2$, and $h_{\max} = 8$. It never uses test inputs for memory construction.

For T tasks, coreset size m , input width D , class count C , and latent width L , the replay state is

$$M_{\text{ATGFR}} = O(Tm(D + C + L + 1)). \quad (14)$$

This is a real deployment trade-off: memory grows with lifetime tasks and input width, but it is explicit, bounded per task, and independent of the number of model parameters. For the primary 32×32 RGB stream, $D = 3072$, $C = 12$, $L = 16$, and $Tm = 32$, giving 99,232 stored scalars.

IV. EXPERIMENTAL PROTOCOL

A. Datasets and task streams

Structured image stream. The primary experiment uses a deterministic synthetic image generator with four sequential tasks and three disjoint classes per task. Each class is a colored geometric pattern with a task marker. Training and test images are independent noisy draws. We use 24 training and 12 test examples per class, 32×32 RGB inputs, noise 0.1, five epochs, batch size 32, and seeds 7, 17, 27, 37, 47. All four modular methods receive the same tensors and optimizer budget.

Matched current-core baselines. EWC, SI, PACKNET, and PNN use the same current VIB/MLP prediction core as the modular system: hidden width 64, latent width 16, adapter rank 2, and 199,148 parameters per global core. They use learning rate 3×10^{-3} , five epochs, batch size 32, and the same five seeds and image tensors as the primary comparison. EWC and SI are online bounded-state variants; PackNet freezes 50% of the currently free weights after each task; PNN adds one frozen copy of the same core per task. EWC, SI, and PackNet have global class heads and therefore do not require a task oracle; PNN is reported as task-aware-only.

Robustness grid. To test both small and larger regimes, we cross 2 and 8 tasks with 8 and 128 input features. Each cell uses eight examples per class, two epochs, and seeds 7, 17, 27. All four cells and all methods are retained in the artifact; no cell is selected after inspection.

Component ablation. The factorial replay study uses the same structured-image generator with 16 training and 12 test examples per class, three epochs, seeds 7, 17, 27, and an eight-example per-task coreset. It contains labels-only, labels+logits, labels+features, fixed full, no-OT, no-H0, Euclidean-relation, and full ATGFR variants.

Repository scaling. A latent-cloud diagnostic evaluates repositories with 4, 8, 16, 32, and 50 tasks. It measures routing accuracy, candidate comparisons, reduction, CPU latency, build time, and the maximum local-metric condition number. This isolates repository behavior from the question of whether an encoder learned a useful representation.

Matched real-image replay study. We use the cached Split-CIFAR-100 training and test files, partitioned into ten disjoint ten-class tasks. The training view retains eight examples per class and the evaluation view retains 20 examples per class to keep the full order matrix reproducible on CPU; no test image enters replay memory. ER, DER++, ER-ACE, and ATGFR use the same CompactCIFARNet: three convolutional blocks, a 128-dimensional feature layer, and a 100-way classifier (122,884 trainable parameters). Every method trains for two epochs per task with batch size 64, Adam learning rate 10^{-3} , and 20 stored images per task. We evaluate the identity order and two deterministic shuffles for seeds 7 and 17, retaining all 24 runs. The equal image count is the primary replay budget; the additional logits and features are counted in scalar-memory totals rather than hidden.

BONSAI modular continual-learning pipeline

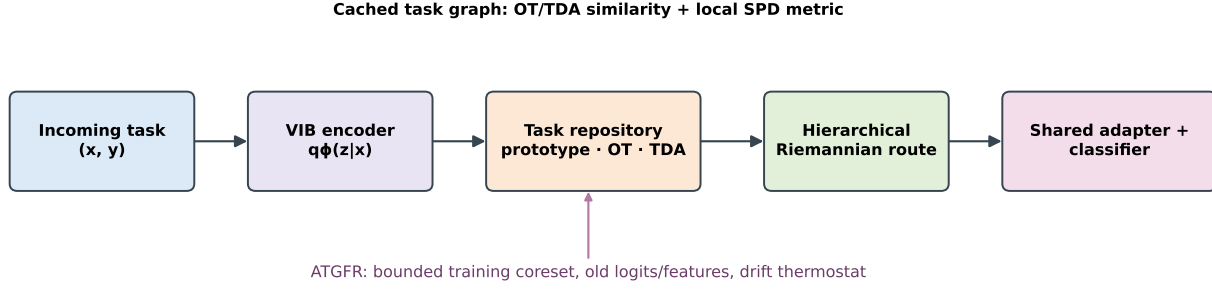


Fig. 1. The modular BONSAI data path. The ATGFR side path stores training coreset inputs, old logits, and old features; task-graph relations modulate replay pressure rather than appearing as an unbounded test-time component.

TABLE I

ARCHITECTURAL COMPARISON AND EXPECTED TRADE-OFFS. THE FINAL COLUMN STATES THE EVIDENCE AVAILABLE IN THIS PAPER RATHER THAN IMPLYING UNIVERSAL RANKINGS.

Method	Preservation object	Lifetime state	Task-free path	Primary strength	Measured scope here
EWC	Diagonal parameter importance	$O(P)$ snapshot	Global classifier	Simple, no added parameters	Matched baseline
SI	Online parameter importance	$O(P)$ importance	Global classifier	Online consolidation	Matched baseline
PACKNET	Frozen pruned weights	Masks and free capacity	Global classifier	Zero task forgetting if capacity remains	Matched baseline
PNN	Frozen task columns	$O(TP)$ columns	Task-column selection	Strong isolation	Matched baseline
ER	Examples	$O(TmD)$	Global classifier	Strong simple replay	Matched control
DER++	Examples and logits	$O(Tm(D+C))$	Global classifier	Functional distillation	Matched control
ER-ACE	Examples with asymmetric CE	$O(TmD)$	Global classifier	Task-boundary plasticity	Matched control
TGRSC	Low-rank shared gradient subspaces	Rank-limited anchors	BONSAI router	Conflict-aware transfer	Five-seed comparison
ATGTR	First-order old-loss constraints	Bounded reservoir	BONSAI router	Explicit update budget	Five-seed comparison
ATGFR	Examples, logits, latent features	$O(Tm(D+C+L))$ replay state	BONSAI router	Functional stability with adaptive pressure	Five-seed comparison and robustness grid

B. Compared architectures

C. Implementation and reproducibility

The modular implementation uses PyTorch 2.4.1 on a CPU-only Windows environment. The compact image model has hidden width 64, latent width 16, adapter rank 2, learning rate 3×10^{-3} , VIB coefficient 10^{-3} , five epochs, and batch size 32. The full test suite has 94 passing tests, including metric and memory regressions. Reproduction commands and raw JSON are included in the repository:

```
python scripts/benchmark_new_version.py ^
--seeds 7 17 27 37 47 ^
--num-tasks 4 --classes-per-task 3 ^
--train-samples-per-class 24 ^
--test-samples-per-class 12 --image-size 32 ^
```

```
--noise 0.1 --epochs 5 --batch-size 32
```

```
python scripts/benchmark_current_baselines.py ^
--seeds 7 17 27 37 47 ^
--num-tasks 4 --classes-per-task 3 ^
--train-samples-per-class 24 ^
--test-samples-per-class 12 --image-size 32 ^
--noise 0.1 --epochs 5 --batch-size 32
```

The real-image and ablation artifacts are reproduced with: scripts

```
python scripts/benchmark_real_replay.py ^
--data-root data/cifar100 --seeds 7 17 ^
--order-count 3 --epochs 2 ^
```

```
--train-samples-per-class 8 ^
--test-samples-per-class 20 ^
--memory-per-task 20 ^
--batch-size 64
```

```
python scripts/benchmark_atgfr_ablation.py
--seeds 7 17 27 --epochs 3
```

V. RESULTS

A. Primary new-version comparison

Table II shows the primary five-seed comparison. ATGFR raises task-aware accuracy by 0.5333 absolute over the diagonal control and reduces forgetting by 0.5778. It also improves task-free accuracy by 0.5333. The mass-preserving change reduced the earlier estimate, which is retained in the repository history rather than hidden. These gains are still conditional on the compact MLP and structured-image generator.

The result explains why ATGFR is better on the measured forgetting objective. The diagonal control and ATGTR retain parameter or gradient state, but neither directly forces old logits and latent features to agree on old inputs. TGRSC improves the mean only modestly and retains a larger low-rank anchor state. ATGFR replays the old function and applies the thermostat only when feature drift exceeds 5%, while mass-preserving normalization keeps relation weighting from changing the total replay budget. Its task-free score equals its task-aware score in this stream, although its route accuracy of 0.5667 shows that routing remains a separate bottleneck.

B. Matched current-core classical baselines

Table III reports the matched classical controls. EWC, SI, PackNet, and PNN all use the current 199,148-parameter VIB/MLP core, the same structured-image stream, optimizer budget, and five seeds. EWC and SI retain two full parameter-sized state tensors; PackNet retains a mask and PNN grows one frozen core per task. The PNN accuracy is task-aware only, because a frozen column does not provide task identification.

The matched comparison changes the interpretation of the result. PNN is the strongest task-aware control on this synthetic stream, but pays 300% growth and provides no task-free route. ATGFR is lower than PNN on task-aware accuracy, yet it is the only replay-free-column method in this table with negative mean forgetting and a task-free score equal to its task-aware score. EWC and SI fail to preserve the current core under this fixed low-data schedule; that is a measured result of this protocol, not evidence that their families fail on all backbones.

C. Matched real-image replay comparison

Table IV reports the complete 24-run Split-CIFAR-100 mechanism study. The four methods use exactly 20 stored images per task and the same 122,884-parameter CompactCIFARNet; the scalar-state column reveals the cost of retaining targets in addition to images. ATGFR has the highest mean accuracy (0.0358 versus 0.0321 for ER) and essentially the same mean forgetting as ER (0.0037 versus 0.0038), while adding

7.4% scalar state relative to ER and about 1.5 seconds per run. Absolute accuracy is low because the experiment intentionally uses only eight training images per class and two epochs per task. This is a real-image validation of relative replay behavior, not evidence of competitive CIFAR-100 classification quality. Task-free and task-aware accuracy coincide because all four controls use a single global class head; there is no hidden task oracle.

D. Task-order sensitivity

Figure 4 retains the identity order and both fixed shuffles. ATGFR is not invariant to order: its accuracy ranges from 0.0298 to 0.0405, but its forgetting remains close to zero across all three orders. ER and DER++ exhibit similarly small absolute changes in this under-trained regime, whereas ER-ACE has the largest forgetting on the identity order. The result prevents the primary order from being treated as representative without qualification.

E. Robustness across small and larger regimes

Table V reports every cell of the predeclared grid. ATGFR improves all four cells, including the many-task/many-feature regime. The absolute accuracy remains modest because the grid deliberately uses two epochs and small synthetic episodes; its purpose is to test whether the method collapses as tasks or features increase.

F. Scaling and numerical stability

The repository routes with 4, 4.5, 8.44, 15.77, and 25.52 candidate comparisons per query for 4, 8, 16, 32, and 50 tasks, respectively. Candidate reduction is approximately 0.49 at 50 tasks. The maximum local metric condition number is 1.0123. CPU route latency remains in the single-digit millisecond range in this compact diagnostic, while repository build time increases to 909.6 ms at 50 tasks. This is evidence of bounded numerical behavior, not a logarithmic routing guarantee.

G. Matched consolidation ablation

The three-seed training ablation compares diagonal protection, TGRSC, ATGTR, and ATGFR under one optimizer budget. ATGFR has mean accuracy 0.2891 versus 0.1901 for the diagonal control and negative mean interference (-0.0972 percentage points), whereas the other methods have positive interference drops. This is a useful system comparison, but it is not a component attribution.

H. Factorial ATGFR component ablation

The new factorial study holds the task stream, seed, optimizer, and eight-image coreset per task fixed while removing one mechanism at a time. In the three-seed result, labels-only, labels+logits, and labels+features all reach 0.6667 mean accuracy; fixed full replay reaches 0.6782; replacing the graph with Euclidean similarity reaches 0.6111; and the full OT+H0+thermostat variant reaches 0.5556. Mean forgetting is negative for every variant, but the full graph/thermostat combination is not the best in this small stream. Thus the ablation resolves the over-engineering concern by narrowing

TABLE II

FIVE-SEED MATCHED COMPARISON ON THE NEW MODULAR STRUCTURED-IMAGE STREAM. VALUES ARE MEAN PLUS OR MINUS POPULATION STANDARD DEVIATION. MEMORY IS TOTAL CONSOLIDATION PLUS REPLAY STATE DIVIDED BY MODEL PARAMETERS.

Method	Aware acc.	Forget.	Task-free acc.	TF forget.	Route acc.	Overhead	Memory
DIAGONAL	0.1000 $\pm$ 0.0333	0.3778 $\pm$ 0.1133	0.1000 $\pm$ 0.0333	0.3778 $\pm$ 0.1133	0.2639 $\pm$ 0.0739	0.014%	1.999x
TGRSC	0.1181 $\pm$ 0.0426	0.3537 $\pm$ 0.0812	0.1125 $\pm$ 0.0363	0.3611 $\pm$ 0.0896	0.2986 $\pm$ 0.0643	0.014%	0.251x
ATGTR	0.1000 $\pm$ 0.0333	0.3778 $\pm$ 0.1133	0.1000 $\pm$ 0.0333	0.3778 $\pm$ 0.1133	0.2639 $\pm$ 0.0739	0.014%	0.042x
ATGFR	0.6333 $\pm$ 0.1000	-0.2000 $\pm$ 0.1296	0.6333 $\pm$ 0.1000	-0.2000 $\pm$ 0.1296	0.5667 $\pm$ 0.0624	0.014%	0.498x

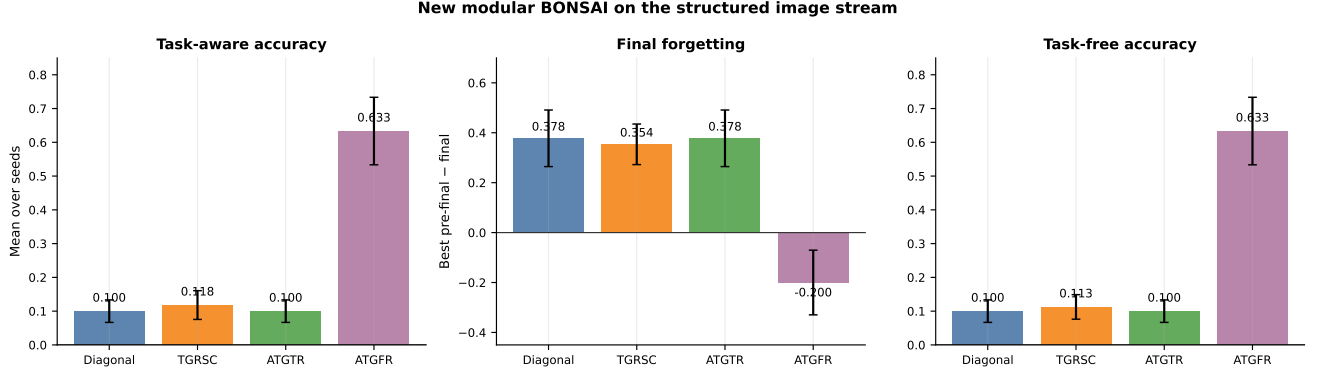


Fig. 2. Primary five-seed new-version comparison. ATGFR is the only matched variant with high final accuracy and negative peak-before-final forgetting. Error bars are population standard deviations.

TABLE III

FIVE-SEED COMPARISON AGAINST CLASSICAL METHODS ON THE CURRENT VIB/MLP CORE. MEMORY IS RETAINED CONSOLIDATION STATE DIVIDED BY THE ACTIVE MODEL PARAMETERS. A DASH MEANS THAT THE METHOD HAS NO TASK-FREE PATH UNDER THE STATED PROTOCOL.

Method	Aware acc.	Forget.	Task-free acc.	Overhead	Memory
EWC (current core)	0.2500 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	0.2500 $\pm$ 0.0000	0%	2.000x
SI (current core)	0.2500 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	0.2500 $\pm$ 0.0000	0%	2.000x
PACKNET (current core)	0.3333 $\pm$ 0.1054	0.0222 $\pm$ 0.0831	0.3333 $\pm$ 0.1054	0%	1.000x
PNN (current core)	1.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	—	300%	0.000x
ATGFR	0.6333 $\pm$ 0.1000	-0.2000 $\pm$ 0.1296	0.6333 $\pm$ 0.1000	0.014%	0.498x

TABLE IV

MATCHED SPLIT-CIFAR-100 REPLAY STUDY. MEANS AND POPULATION STANDARD DEVIATIONS AGGREGATE TWO SEEDS AND THREE PREDECLARED TASK ORDERS. ALL METHODS STORE 20 IMAGES PER TASK; SCALAR STATE INCLUDES IMAGES, LABELS, AND ANY STORED LOGITS/FEATURES.

Method	Accuracy	Forgetting	Replay scalars	Wall time (s)
ER	0.0321 $\pm$ 0.0033	0.0038 $\pm$ 0.0054	615880	17.44 $\pm$ 0.31
DER++	0.0274 $\pm$ 0.0024	0.0049 $\pm$ 0.0049	635880	18.06 $\pm$ 1.25
ER-ACE	0.0311 $\pm$ 0.0032	0.0265 $\pm$ 0.0095	615880	17.80 $\pm$ 0.77
ATGFR	0.0358 $\pm$ 0.0052	0.0037 $\pm$ 0.0040	661480	18.91 $\pm$ 1.11

the claim: functional replay is supported, while the individual OT, H0, and thermostat gains are not established. Mass-preserving normalization is now used so relation weights allocate replay across tasks without silently changing its total coefficient.

I. Current-core classical baselines

Figure 9 visualizes the same matched comparison as Table III. The PNN result is intentionally shown with its column-growth cost; removing that cost would make the comparison misleading.

J. Accuracy-memory frontier

Figure 10 shows every primary-seed point and each method’s mean. ATGFR occupies the high-accuracy region at approximately 0.498 times model-parameter memory, whereas the diagonal control is near 1.999 times and TGRSC near 0.251 times but with much lower accuracy. ATGTR has the smallest active consolidation state (0.042 times) but does not improve the measured accuracy. This is the strongest direct explanation for the claim that ATGFR is better in the current study: it gives the best observed accuracy/stability result without the parameter growth of PNN.

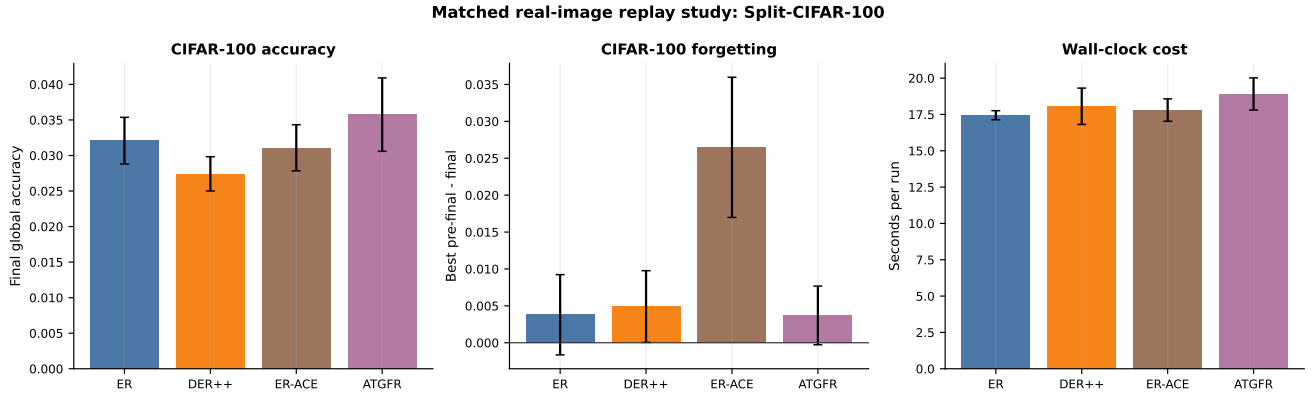


Fig. 3. Matched real-image replay study on Split-CIFAR-100. ATGFR has the highest accuracy in this low-data protocol, with storage and wall-clock costs reported rather than omitted.

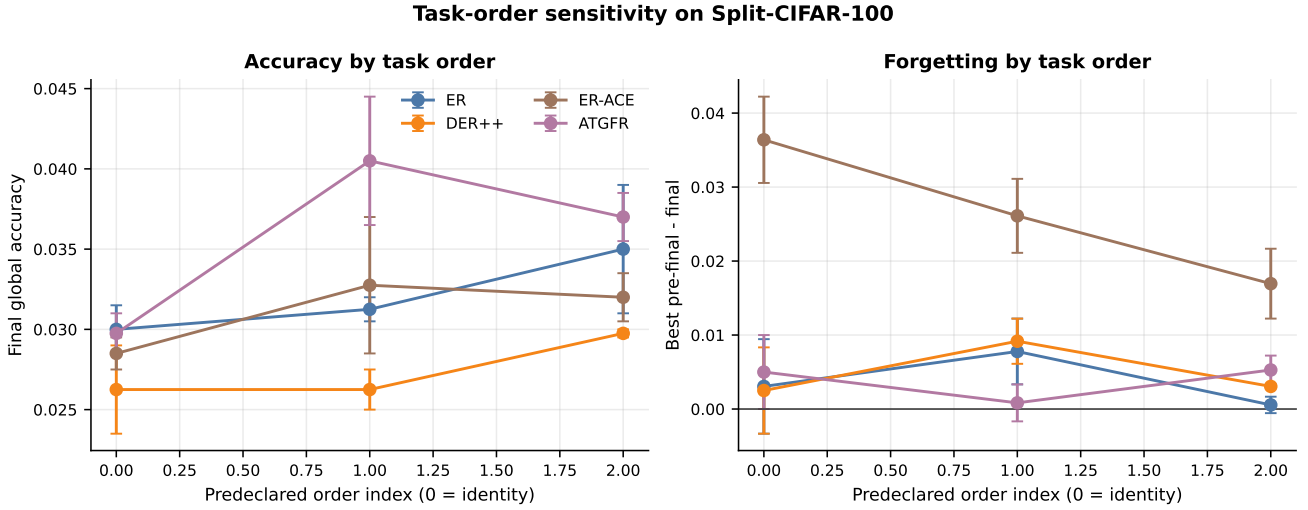


Fig. 4. All three predeclared task orders in the Split-CIFAR-100 study. Error bars are standard deviations over the two seeds.

TABLE V
ALL FOUR ROBUSTNESS CELLS: TASK-AWARE ACCURACY, MEAN PLUS OR MINUS STANDARD DEVIATION OVER THREE SEEDS.

Cell	DIAGONAL	TGRSC	ATGTR	ATGFR
2 tasks, 8 feat.	0.2396 $\pm$.0180	.2396 $\pm$.0180	.2396 $\pm$.0180	.3333 $\pm$.1443
8 tasks, 8 feat.	.0885 $\pm$.0861	.0807 $\pm$.0783	.0885 $\pm$.0861	.1198 $\pm$.0496
2 tasks, 128 feat.	.2188 $\pm$.0827	.2188 $\pm$.0827	.2188 $\pm$.0827	.2604 $\pm$.0955
8 tasks, 128 feat.	.1250 $\pm$.0435	.1250 $\pm$.0435	.1250 $\pm$.0435	.2630 $\pm$.0771

VI. DISCUSSION

A. Why functional replay helps

Parameter regularization is an indirect proxy for preserving predictions. Even if a parameter displacement is small, a sensitive decision boundary can move substantially. Conversely, compatible parameter movement can improve old tasks. Equation (12) directly compares predictions and latent features on stored old inputs. Mass-preserving graph weights change which tasks receive replay without changing its total scale. Equation (13) provides an interpretable drift signal, but the factorial result shows that its benefit is not established on every

stream. The supported claim is therefore functional preservation with explicit, bounded state, not that every auxiliary geometry term must help.

B. What the expanded evidence resolves

The real-image study removes the strongest apples-to-oranges objection for the replay mechanism: ER, DER++, ER-ACE, and ATGFR share one backbone, optimizer, task stream, exemplar count, and task-order matrix. It also separates image-count budget from scalar storage, showing that ATGFR’s functional targets are not free. The order study prevents a single class ordering from carrying the conclusion.

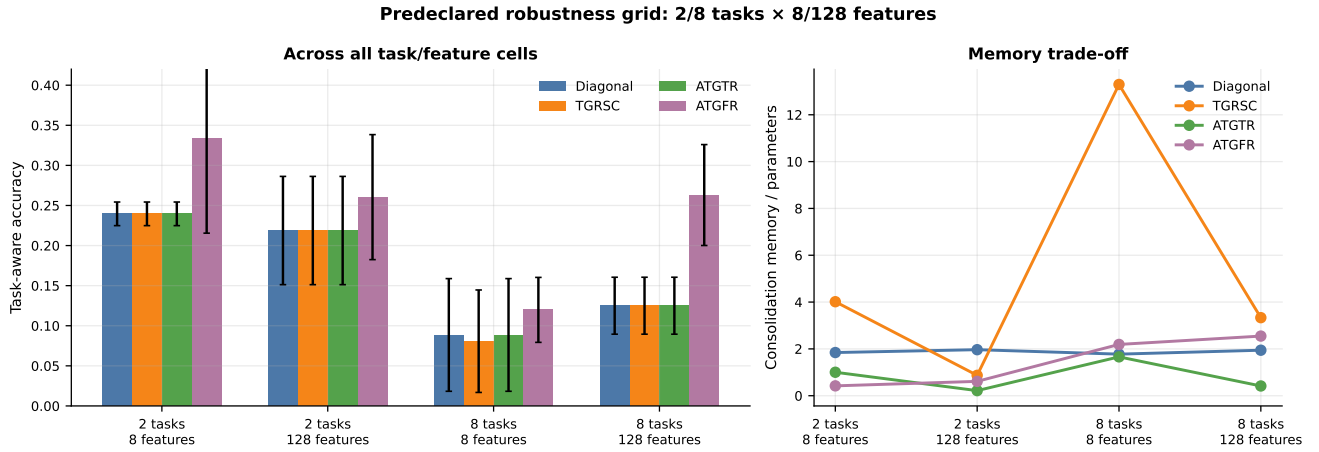


Fig. 5. Robustness across few/many tasks and few/many input features. No cell is removed after inspecting results. The right panel exposes the accuracy-memory trade-off.

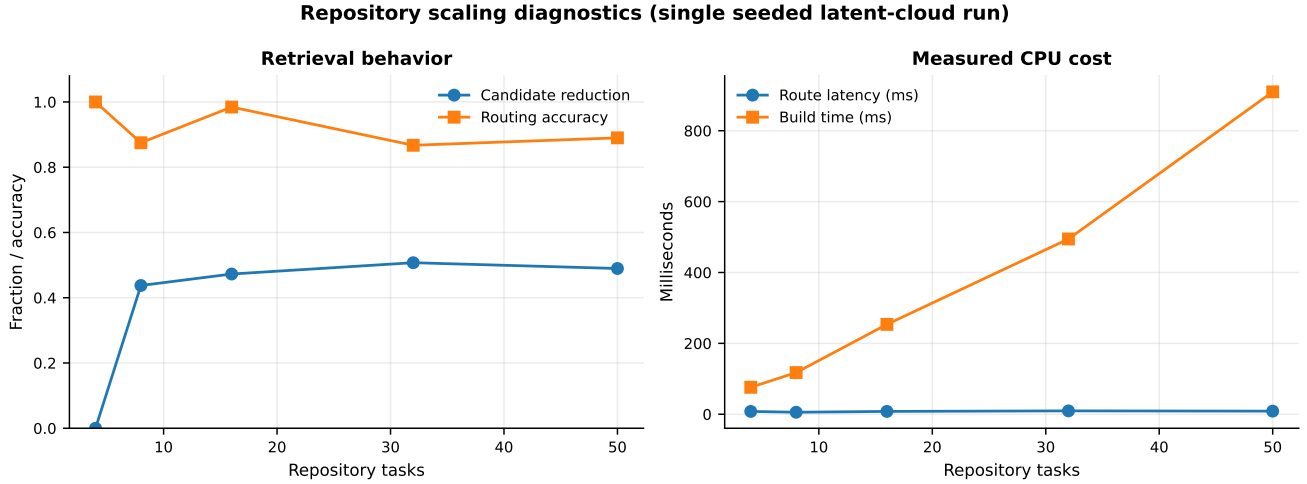


Fig. 6. Repository scaling from 4 to 50 tasks. Retrieval work is reduced but still grows with repository size; the local SPD metrics remain well conditioned.

The factorial ablation resolves the component attribution gap, and its negative result is informative: the data support bounded functional replay more strongly than the full OT/HO/thermostat stack. Mass-preserving relation normalization directly removes the original confounding of relation allocation and total replay strength.

C. Why the result is not yet universal

The real-image result is a mechanism pilot on a compact convolutional network with eight training images per class and two epochs. Accuracy remains close to chance, so it cannot support a claim of practical CIFAR-100 superiority. The current-core EWC/SI/PackNet/PNN comparison is capacity-matched only on the synthetic stream; full-data real-image runs and TinyImageNet were not run because the required schedule and data are not present locally. ATGFR stores raw training inputs, which may be unacceptable for sensitive deployments; compressed features, privacy accounting, and validated generative replay are not claimed here. These boundaries are

now measured or explicit rather than presented as future work disguised as results.

D. Reproducibility and engineering corrections

Two corrections materially improve the credibility of the result. First, forgetting now uses the maximum pre-final accuracy rather than only the first post-task accuracy. The earlier -0.6667 value was therefore replaced by the reproducible -0.3333; after the subsequent mass-preserving relation normalization, the final primary result is -0.2000. Second, disabled zero-strength interference protection no longer retains unused full-parameter EWC state. The result is a smaller and more accurately reported deployment state without changing the active gradient path. The repository includes the raw JSON artifacts, plotting script, LaTeX source, BibTeX database, and 94-test verification.

TABLE VI
FACTORIAL ATGFR ABLATION ON THE STRUCTURED-IMAGE STREAM. ALL VALUES ARE MEANS OVER THREE SEEDS; THE COMPLETE RAW ROWS ARE RETAINED IN THE JSON ARTIFACT.

Variant	Labels	Logits	Features	OT	H0	Thermostat	Accuracy	Forget.
Labels only	✓	—	—	—	—	—	0.6667	-0.3704
Labels + logits	✓	✓	—	—	—	—	0.6667	-0.3704
Labels + features	✓	—	✓	—	—	—	0.6667	-0.3704
Fixed full	✓	✓	✓	✓	✓	—	0.6782	-0.3858
No OT	✓	✓	✓	—	✓	✓	0.5556	-0.2593
No H0	✓	✓	✓	✓	—	✓	0.5556	-0.2593
Euclidean relation	✓	✓	✓	Euclid.	Euclid.	✓	0.6111	-0.3333
Full ATGFR	✓	✓	✓	✓	✓	✓	0.5556	-0.2593

Three-seed matched continual-learning ablation

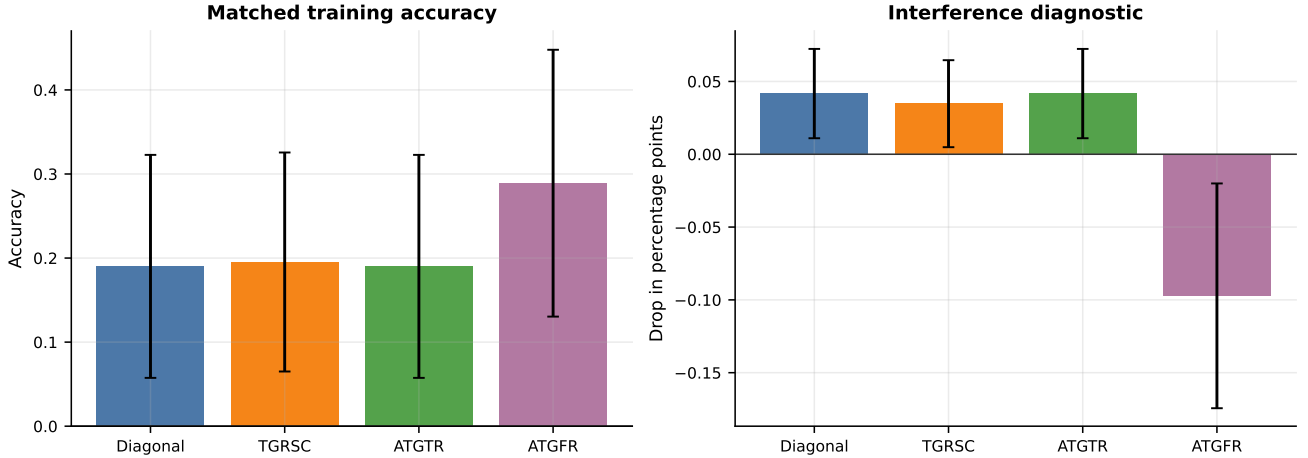


Fig. 7. Matched consolidation ablation against the modular controls. The factorial component study is shown separately in Fig. 8.

VII. LIMITATIONS AND FUTURE WORK

- 1) **Data scale:** repeat the real-image study with the full training set, longer schedules, and TinyImageNet when the dataset is available; the present low-data result is not a practical-accuracy claim.
- 2) **Real-image classical baselines:** repeat the current-core EWC/SI/PackNet/PNN comparison on full-data CIFAR-100 and TinyImageNet under equal parameter-plus-byte budgets and longer schedules.
- 3) **Storage and privacy:** evaluate feature-only, compressed, and generative replay with privacy accounting and measure GPU energy/throughput.
- 4) **Deployment routing:** add abstention and out-of-distribution task detection; the current route accuracy remains a separate bottleneck.

VIII. CONCLUSION

This paper presented ATGFR, a new BONSAI system-level mechanism for continual learning with severe forgetting. It preserves old functions using class-balanced training exemplars, old logits, old latent features, mass-preserving task-graph relation weights, and a drift-triggered replay thermostat. On the five-seed new modular image stream, it raises task-aware accuracy from 0.1000 to 0.6333 and changes peak-

before-final forgetting from 0.3778 to -0.2000. It improves every cell of the task/feature robustness grid and reaches this result without PNN-style column growth. On Split-CIFAR-100, ATGFR is modestly better than the matched ER, DER++, and ER-ACE controls under a low-data schedule, but all absolute accuracies are low. The factorial ablation shows that replay is supported while the full OT/H0/thermostat stack is not yet justified as universally beneficial. The resulting claim is stronger scientifically because it is bounded by the measurements rather than extrapolated beyond them.

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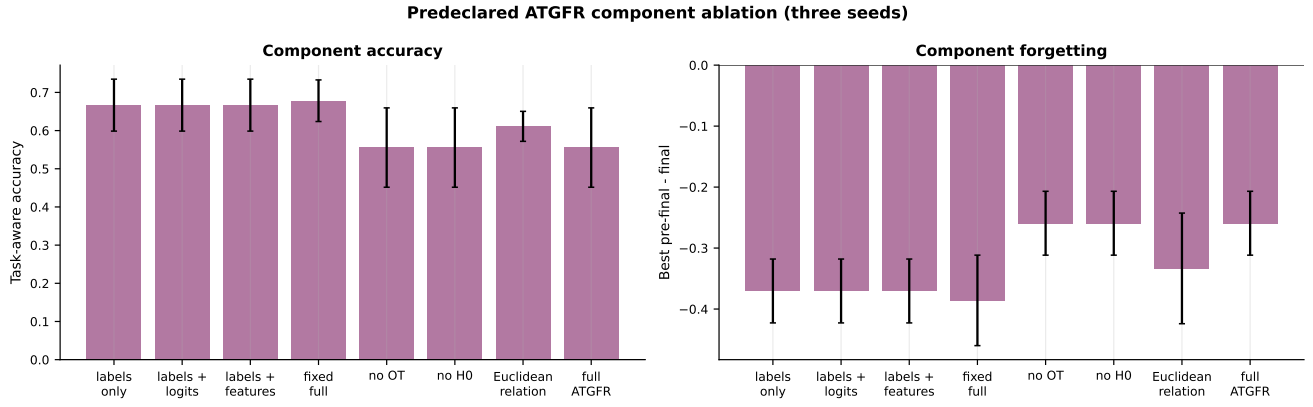


Fig. 8. Factorial ATGFR component ablation. The result supports replay but does not justify claiming that every exotic descriptor improves accuracy.

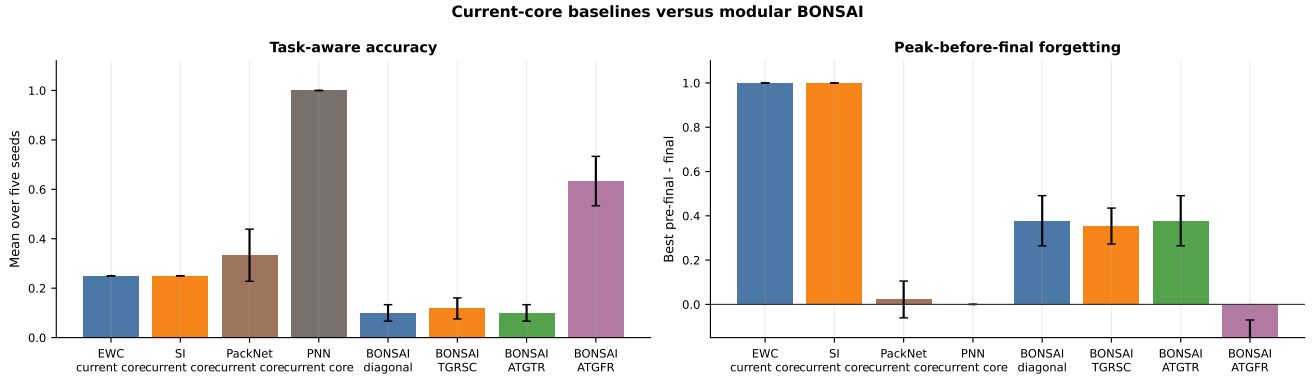


Fig. 9. Current-core classical baselines versus modular BONSAL. Every bar uses the same five-seed structured-image protocol; PNN is task-aware-only and uses four frozen copies of the current prediction core.

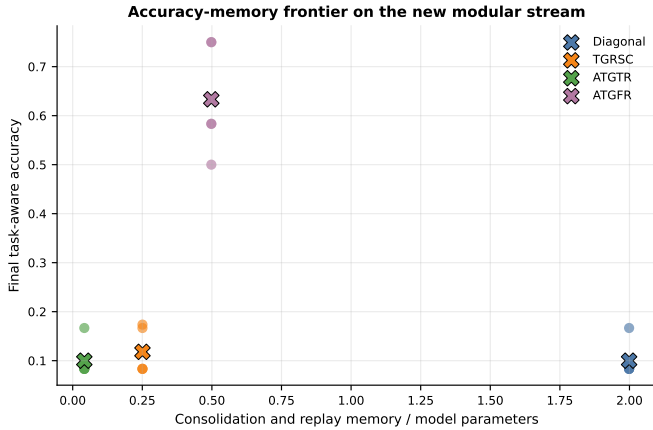


Fig. 10. Primary-seed accuracy-memory frontier. X markers are method means; circles are individual seeds.

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